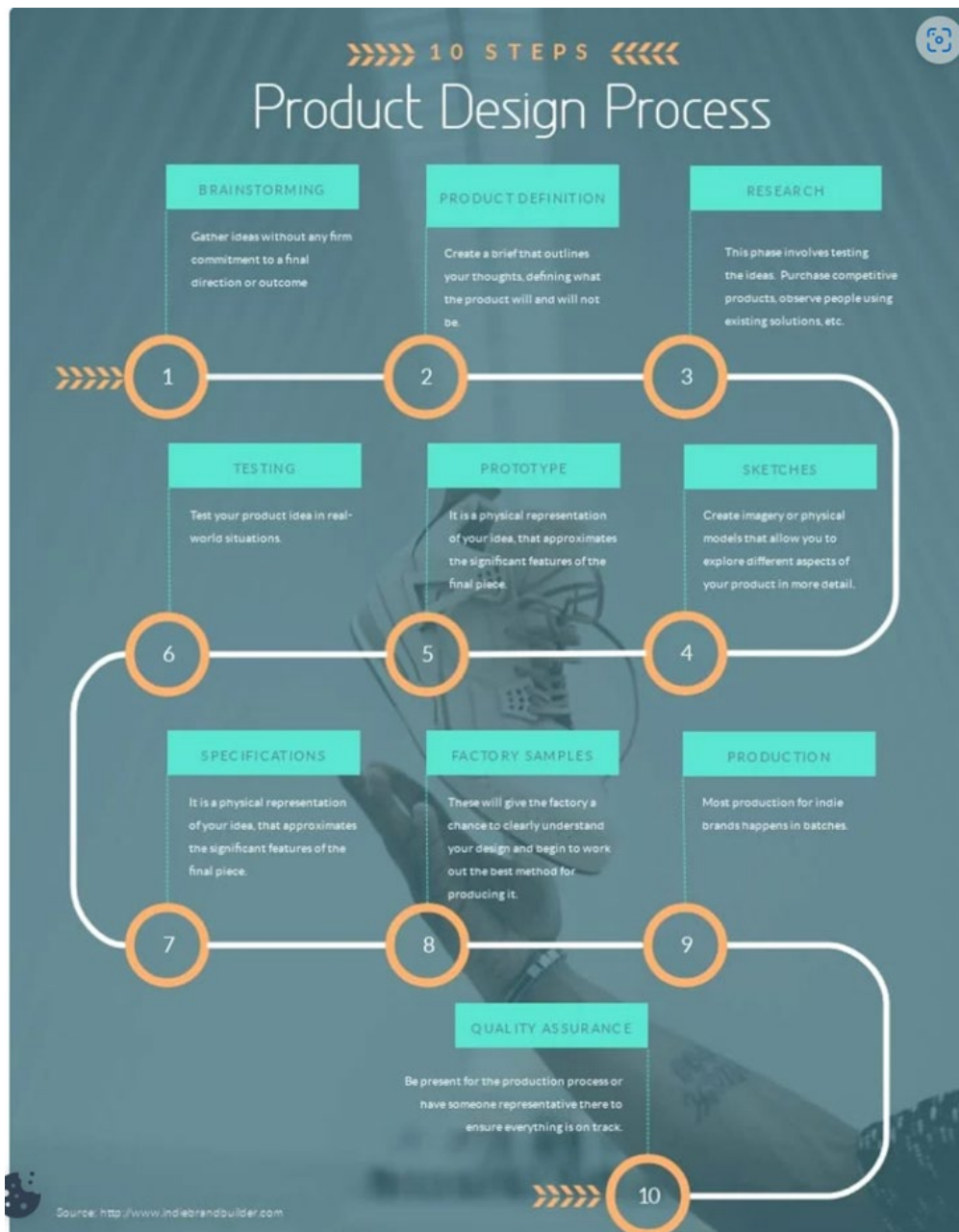
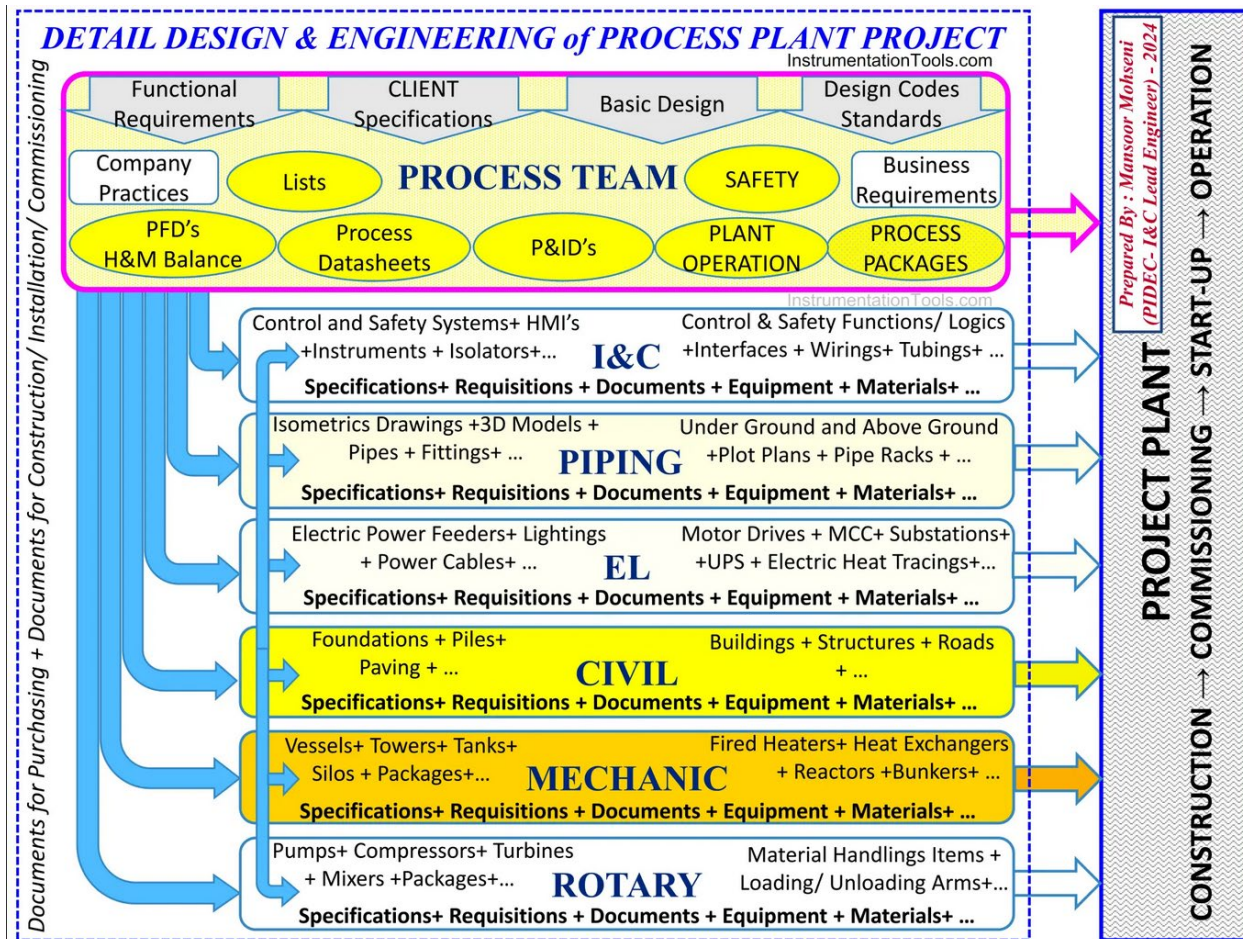


Lab 3 Supplemental – Engineering Design Process Documentation

Lab- 2 – Connection:

As a part of product design process of Lab-2, you have already gone through the basic engineering design thinking process till prototyping and testing phase. Next step is to think about how to set up a production facility by following the steps outlined in the following process flow diagram and the project life cycle diagram (on next page). Watch the following video to initiate the thought process - [3DEXPERIENCE Delmia automated bridge manufacturing factory](#)





As shown in detail design & engineering of process plant project diagram above, creating a production facility will need engineering documentation created by various disciplines of engineering (such as I &C, Piping, Electrical, Civil, Mechanical & Rotary) in terms of specifications, engineering drawings (such as layout drawing, wiring drawings, piping isometrics, etc.) as well as other reports (such as HSE and HAZOP reports). All these documents and information required for building the production facility will be provided to a project scheduler to create a project schedule.

Following are some example templates of various documents you will be creating as part of this lab.

[illegible]

Notch Cuts

30 x 12 x 12 x 1 x

3/8" 2.5" 1/2" 3/4" 3/8"

A B C D

4 1/8" 3 3/8" 4 1/8" 4.5"

Wall Cuts

1.75" 6 x 1.75"

A J J

6 x 6 x

3.75" 12 x

C K

12 x

[Lincoln Sticks: Popsicle Stick Log Cabins : 8 Steps \(with Pictures\) - Instructables](#)

Follow the task numbering system as steps highlighted in yellow.

Detailed Task List:

Task 1- Preparation:

- **Sub Task 11- Gather Materials:**
- **Sub Task 12- Design Planning:**
 - Sub Sub task 121 - Determine the bridge's desired span (length)
 - Sub Sub task 122 - Sketch different bridge designs (consider truss, beam, or arch)
 - Sub Sub task 123 - Calculate the required number of popsicle sticks based on the design

Task 2 - Construction:

- **Sub Task 21 - Cutting and Shaping:**
 - Cut popsicle sticks to the desired lengths according to your design
 - If needed, carefully trim or shape popsicle sticks to fit specific angles
- **Sub Task 22 - Assembling the Structure:**
 - Start by creating small triangular units (key for strength)
 - Glue popsicle sticks together to form the main structure of the bridge, following your design
 - Gradually build up the bridge, adding layers and reinforcing critical connection points with additional glue
- **Sub Task 23- Reinforcement:**
 - Add extra layers of popsicle sticks to key areas like the base and support beams
 - Consider using small pieces of cardboard or construction paper to reinforce specific sections

Testing and Refinement:

- **Load Testing:**
 - Place the bridge on supports (like two chairs) and gradually add weight (books, small objects) to test its load-bearing capacity
 - Observe where the bridge shows signs of stress or weakness
- **Adjustments:**
 - If needed, reinforce weak areas with additional glue and popsicle sticks
 - Modify the design based on the test results

- Open Microsoft Project and open the example *.mpp file uploaded on FOL. Fill in the tasks and timeline (duration) in the mpp file and create Gantt chart.



Company name: Assessment carried out by:
Date of next review: Date assessment was carried out:

[illegible]

Rubric for ENGR-3027 Lab 3 Report

- 1) Specification sheets – 1 mark
- 2) Project Schedule – 1 mark
- 3) Production Facility Layout Drawing – 1 mark
- 4) HSE report – 1 mark
- 5) Bridge making instructions or tasks list – 1 mark